

Course Name	Fundamentals of Fog & Edge Computing	Date	03-01-2025
Course Code	BCSE313L	SLOT	F1 & F2
		Class No.	VL2024250501570/1575

Tutorial Sheet 1

Problem Statement 1

A large-scale agricultural company is looking to implement a smart farming system. The system will collect data from various sensors placed throughout their fields, including soil moisture sensors, weather stations, and drone-mounted cameras. This data will be used to optimize irrigation, fertilization, and pest control. Following tasks are carried out by the company:

- Real-time analysis of sensor data to trigger immediate irrigation adjustments.
- Long-term storage and analysis of historical weather patterns to predict future crop yields.
- On-device processing of images captured by drones to identify and count individual plants.
- Centralized management and control of the entire smart farming system, including software updates and security management.

Identify which type of computing (Mist, Fog, Edge, or Cloud etc.) would be most suitable for each task and explain why.

Problem Statement 2

A major metropolitan area is planning to implement a large-scale network of autonomous vehicles for public transportation. This network will involve hundreds of self-driving buses and cars operating across the city.

Identify specific tasks within this public transportation system that would be best suited for each of the following computing paradigms:

- Mist Computing
- Edge Computing
- Fog Computing
- Cloud Computing

Explain the rationale behind your choices, considering factors like latency, bandwidth, data processing needs, and security.

Problem Statement 3

For the given scenario in Problem Statement 2 (refer to the autonomous vehicle public transportation scenario), determine the potential entities residing in the various layers

of the fog & edge hierarchy architecture. Additionally, sketch a diagram depicting the layers and the potential entities.

Problem Statement 4

Referring to Problem Statement 1 - smart farming system, illustrate different Business Models?

Problem Statement 5

What advantages of fog and edge computing are achieved in the scenario of smart farming system.